

T09 – Customer RFM Segmentation Tool

Project Report: Customer Segmentation, Campaign Recommendations and Reflection

1. Introduction and Project Objective

The objective of this project was to build a working Customer RFM Segmentation Tool using Recency, Frequency and Monetary (RFM) analysis. The tool classifies customers into actionable segments and recommends a targeted campaign for each segment. The final solution includes a data-analysis notebook, an Excel workbook containing the required outputs, and a Streamlit application for customer-level lookup.

Dataset and Data Quality

The Sales sheet contained 3,000 transaction records and 16 variables, including order ID, customer ID, order date, region, channel, product information, units, price, discount, revenue, cost and gross margin. The actual dataset contained 426 unique customers. Data quality checks found no missing values, no duplicate order IDs, no invalid dates, and no zero or negative revenue orders. The transaction period was 1 April 2025 to 31 March 2026.

2. RFM Methodology

RFM was calculated at customer level. Recency is the number of days between the latest transaction date in the dataset and a customer's most recent purchase. Frequency is the number of orders placed by the customer. Monetary is the total revenue generated by the customer. The three measures were converted into scores from 1 to 5 using quintile-based scoring. Lower Recency receives a better score, while higher Frequency and Monetary values receive better scores.

Scoring and Segmentation

Rank-based qcut scoring was used to avoid problems caused by tied values. The three scores were combined into an RFM score such as 555. Customers were then assigned to eight project-defined segments: Champions, Loyal Customers, Potential Loyalists, New / Promising, At Risk, Hibernating, Lost and Needs Attention. These segmentation rules are analytical rules created for this project rather than universal RFM definitions.

3. Application Development

The Streamlit application provides a simple customer lookup interface. A user enters a customer ID and the application retrieves the customer's Recency, Frequency, Monetary value, individual R/F/M scores, combined RFM score, segment and recommended campaign. An invalid customer ID produces an appropriate error message. The application uses the customer-level CSV generated from the RFM analysis.

4. Results and Segment Analysis

The segmentation produced eight customer groups. Champions represented 75 of the 426 customers (approximately 17.6%) but accounted for 58.53% of historical revenue. Their average Recency was 15.92 days, average Frequency was 22.84 orders and average Monetary value was ₹271,091.74. This indicates a recent and highly valuable group where retention is important.

At Risk customers numbered 51 and contributed 12.86% of historical revenue. Their average Recency was approximately 129.67 days, average Frequency was 5.22 orders and average Monetary value was ₹87,595.00. This group combines meaningful historical value with comparatively long inactivity and is therefore suitable for targeted win-back activity.

Potential Loyalists had average Recency of 14.38 days and average Frequency of 3.38 orders, indicating recent engagement with room to build repeat behaviour. New / Promising customers had average Recency of 17.15 days but average Frequency of 1.60 orders, suggesting an opportunity to encourage a second purchase.

Hibernating customers had average Recency of 138.88 days, while Lost customers had average Recency of 237.42 days. Lost customers also had lower average Monetary value of ₹6,055.90, supporting a lower-cost recovery approach.

Segment	Customers	Total Revenue	Avg Recency	Avg Frequency	Avg Monetary	Revenue Share
Champions	75	₹20,331,880.60	15.92	22.84	₹271,091.74	58.53%
At Risk	51	₹4,467,344.75	129.67	5.22	₹87,595.00	12.86%
Loyal Customers	55	₹3,106,362.70	42.98	7.45	₹56,479.32	8.94%
Hibernating	64	₹2,447,881.55	138.88	2.47	₹38,248.15	7.05%
Needs Attention	54	₹2,201,941.80	62.39	2.94	₹40,776.70	6.34%
Potential Loyalists	52	₹1,552,171.90	14.38	3.38	₹29,849.46	4.47%
Lost	55	₹333,074.35	237.42	1.56	₹6,055.90	0.96%
New / Promising	20	₹294,654.15	17.15	1.60	₹14,732.71	0.85%

5. Campaign Recommendations

The main recommendation is to use differentiated customer treatment rather than a single campaign for the entire customer base. Champions should receive retention-focused benefits such as exclusive rewards, early access and personalized bundles. Loyal Customers can be encouraged toward higher-value purchases through loyalty benefits and personalized bundles. At Risk customers should receive targeted win-back communications because of their historical value and inactivity. Hibernating and Lost customers should receive progressively lower-cost reactivation efforts. Potential Loyalists and New / Promising customers should receive communications designed to create a second or repeat purchase.

Segment	Campaign	Objective	Channel	KPI
Champions	VIP Loyalty Campaign	Retain high-value customers	Email + WhatsApp	Repeat purchase rate
Loyal Customers	Loyalty Upgrade	Increase value and retention	Email + App/WhatsApp	Average order value
At Risk	Win-Back Campaign	Reactivate valuable inactive customers	Email + WhatsApp	Reactivation rate
Hibernating	Come-Back Campaign	Re-engage inactive customers	Email	Re-engagement rate
Needs Attention	Engagement Campaign	Increase purchase frequency	Email + App	Purchase frequency
Potential Loyalists	Second-Purchase Campaign	Convert recent customers into repeat buyers	Email + WhatsApp	Second-purchase rate
New / Promising	Welcome-to-Loyalty Campaign	Encourage first repeat purchase	Email + App	Repeat purchase rate
Lost	Last-Chance Win-Back	Attempt low-cost customer recovery	Email	Reactivation rate

6. Verification and Testing

The solution was verified at both the analysis and application levels. Selected customers were compared with their underlying Sales transactions. For example, CUST0387 was checked against the source transactions and had 514 orders, latest purchase date 31 March 2026, total revenue of ₹6,389,543.50 and Recency of 0 days. Its R/F/M scores were 5/5/5 and it was classified as Champions. The application was also tested with an invalid customer ID to confirm appropriate error handling.

7. Business Recommendation

The business should prioritize retention of Champions while using targeted win-back activity for At Risk customers. Champions generated the largest historical revenue share, while At Risk customers have meaningful historical monetary value despite being inactive. Potential Loyalists and New / Promising customers should be encouraged toward repeat purchases. Hibernating and Lost customers can be addressed through lower-cost reactivation campaigns, with marketing effort reduced when customers do not respond.

8. AI Use, Reflection and Learning

AI assistance was used during the project to understand the assignment, generate and refine Python code, handle a qcut scoring issue, develop segmentation rules, design campaign ideas, create the Streamlit application and troubleshoot deployment. The AI Use Log records the project interactions and the changes made after reviewing outputs. The analysis was not accepted blindly: code was executed in Colab, selected customers were checked against the underlying transactions, and the application was tested with both valid and invalid customer IDs.

9. Limitations and Future Improvements

The segmentation rules are project-specific and should be treated as a decision-support framework rather than a permanent definition of customer behaviour. The dataset is synthetic, and RFM uses historical transaction behaviour without additional factors such as customer satisfaction, product profitability by customer, campaign response or future purchase probability. A future version could combine RFM with customer lifetime value, product preferences and campaign-response data.

10. Conclusion

The project produced an end-to-end RFM segmentation solution: transaction data was transformed into customer-level RFM measures, customers were segmented, campaign actions were assigned, results were verified, and the analysis was implemented in a customer lookup application. The notebook, workbook and Streamlit app together provide both analytical evidence and a practical interface for using the results.